Work, energy and power

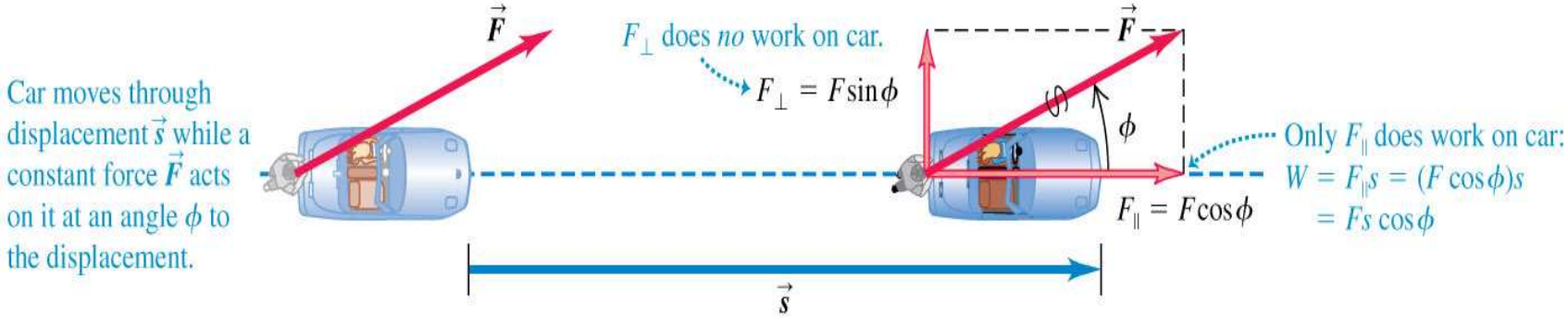
An alternative approach to mechanics

- Many problems in mechanics involve forces which are not constant.
- Solving this type of problems using Newton's laws can be very difficult.
- An alternate method based on the concepts of work, energy and power can greatly simplify calculations.

Work done by a constant force

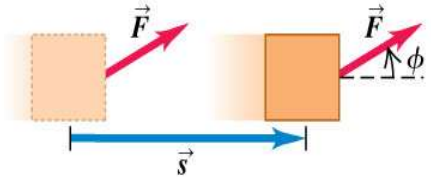
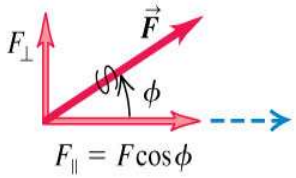
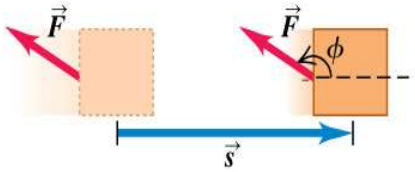
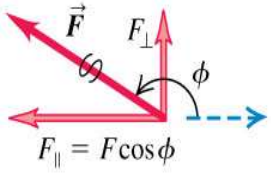
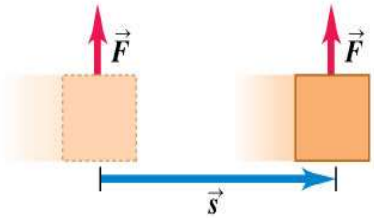
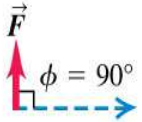
- The work done by a constant force is defined as $W = \vec{F} \cdot \vec{s} = Fs\cos\phi$

  where ϕ is the angle between the force and the displacement.



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Work can be positive, negative or zero

Direction of Force (or Force Component)	Situation	Force Diagram
(a) **Force $\vec{F}$ has a component in direction of displacement:** $W = F_{\parallel}s = (F\cos\phi)s$ Work is *positive*.		
(b) **Force $\vec{F}$ has a component opposite to direction of displacement:** $W = F_{\parallel}s = (F\cos\phi)s$ Work is *negative* (because $F\cos\phi$ is negative for $90° < \phi < 180°$).		
(c) **Force $\vec{F}$ (or force component $F_\perp$) is perpendicular to direction of displacement:** The force (or force component) does *no* work on the object.		

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Example 1

The diagram shows a force-displacement graph of an object. The force acts in the direction of the displacement for the first 5 m and opposite to the displacement in the next 5 m. What is the work done for the whole journey?

For the first 5 m, $\phi = 0^{\circ}$.

$W = \vec{F} \cdot \vec{s} = Fs\cos 0^{\circ}$

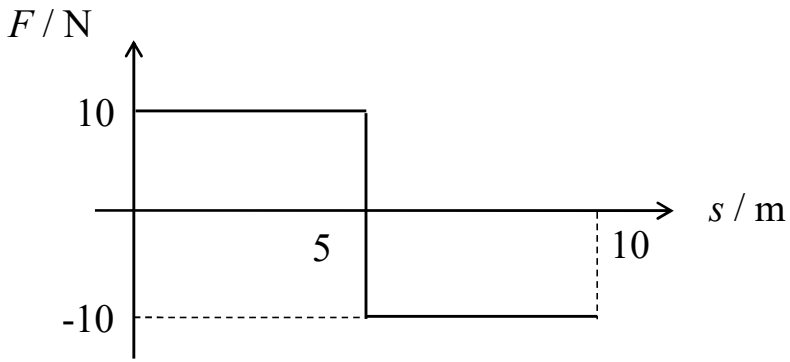
$= 10 \times 5 = 50$ J

For the next 5 m, $\phi = 180^{\circ}$.

$W = \vec{F} \cdot \vec{s} = Fs\cos 180^{\circ}$

$= -10 \times 5 = -50$ J

Total work done is zero.



Example 2

a) Steve exerts a steady force of magnitude 210 N on a stalled car so that it moves a distance of 18 m. The car has a flat tire. So to make the car track straight Steve must push at an angle of 30° to the direction of motion. How much work does Steve do?

b) Steve pushes another stalled car with a steady force $\vec{F} = 160\hat{i} - 40\hat{j}$ N. The displacement of the car is $\vec{s} = 14\hat{i} + 11\hat{j}$ m. How much work does Steve do?

a) $W = \vec{F} \cdot \vec{s} = Fs\cos 30^{\circ}$

$= 210 \times 18\cos 30^{\circ} = 3.3$ kJ

b) $W = \vec{F} \cdot \vec{s} = (160\hat{i} - 40\hat{j}) \cdot (14\hat{i} + 11\hat{j})$

$= 160 \times 14 - 40 \times 11$

$= 1.8$ kJ

Work due to many forces

If there are many forces acting on an object, we can use one of the following two approaches to calculate the total work :

i) Compute the work done by each separate force and add them together.

ii) Compute the net force (vector sum) and find the work done by the net force.

Energy

- Energy is defined as the ability to do work.
- Hence, energy and work are equivalent.
- The SI unit of energy is the joule (J).
- The different types of energy are light, heat, sound, electrical, chemical and mechanical.
- Mechanical energy comprises kinetic energy and potential energy.

Principle of conservation of energy

- The principle of conservation of energy states that energy
  - can neither be created nor destroyed in any process.
  - can be converted from one type to another (including work).
  - can be transferred from one body to another with the total amount remaining constant.
- The total energy of an isolated system (one that is not interacting with other systems) is constant.

Total Work

- Suppose a constant **net force** F acting on a particle of mass m along the $+x$-axis results in a displacement s along the $+x$-axis such that it has an initial velocity v_1 and final velocity v_2. We apply the following kinematical equation.

$$v_2^{\,2} = v_1^{\,2} + 2a_x s \qquad \text{or} \qquad a_x = \frac{v_2^{\,2} - v_1^{\,2}}{2s}$$

Multiplying the above equation with m

$$ma_x = m\frac{v_2^{\,2} - v_1^{\,2}}{2s}$$

$$F = m\frac{v_2^{\,2} - v_1^{\,2}}{2s} \qquad \text{since} \quad F = ma_x$$

$$Fs = \frac{1}{2}m v_2^{\,2} - \frac{1}{2}m v_1^{\,2}$$

$$W_{total} = \frac{1}{2}m v_2^{\,2} - \frac{1}{2}m v_1^{\,2} \qquad \text{since} \quad W_{total} = Fs$$

Kinetic energy and Work-energy theorem

- The quantity $\frac{1}{2}mv^2$ is known as kinetic energy (KE or K).
- The SI unit of kinetic energy is joule (J).
- The work-energy theorem states that the work done by the net force on an object equals the change in the kinetic energy of the object.
- Kinetic energy cannot be negative because of the factor v^2.
- If the object at rest (i.e. $v = 0$), then its kinetic energy is 0.
- Note that the direction in which the object moved is not considered in the calculation of its kinetic energy.

Work-energy theorem

- From $W_{\text{total}} = K_2 - K_1 = \Delta K$,

  if $W_{\text{total}} > 0$, then $K_2 > K_1$

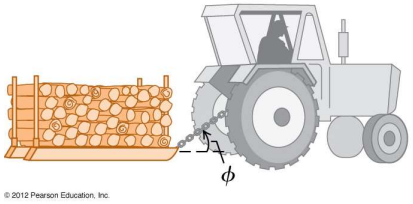
  if $W_{\text{total}} < 0$, then $K_2 < K_1$

  if $W_{\text{total}} = 0$, then $K_2 = K_1$

- The work-energy theorem applies even when the forces are not constant and the particle's trajectory is curved.

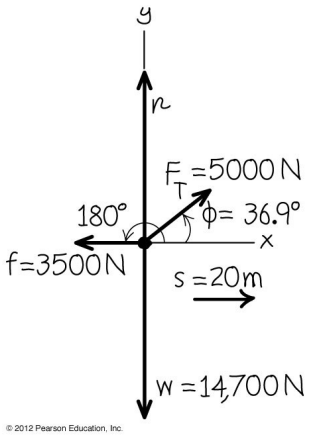
Example 3

A farmer hitches his tractor to a sled loaded with firewood and pulls it a distance of 20 m along level ground (see figure). The total weight of the sled and load is 14700 N. The tractor exerts a constant force of 5000 N at an angle of 36.9° above the horizontal. A friction force of 3500 N opposes the sled's motion. (a) Find the work done by each force acting on the sled and (b) the total work done by all the forces. (c) Suppose the sled's initial speed is 2.0 m/s. What is the speed of the sled after it moves 20.0 m?



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(b) Free-body diagram for sled



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(a) $W_{\text{weight}} = W_{\text{normal}} = 0$ (angle between force and s is 90°)

$W_{\text{friction}} = 3500 \times 20 \times \cos 180^\circ = -70 \times 10^3$ J

$W_{\text{tractor}} = 5000 \times 20 \times \cos 36.9^\circ = 80 \times 10^3$ J

(b) $W_{\text{total}} = 0 + 0 + 80 \text{ kJ} + -70 \text{ kJ} = 10$ kJ

Weight = 14700, hence $m = 14700/9.80 = 1500$ kg

(c) W_{total} = change in KE (work-energy theorem).

$10000 = \frac{1}{2}(1500)v^2 - \frac{1}{2}(1500)2^2$

Hence $v = 4.2$ m/s

Work done by non-constant force

- If the force is not constant, the infinitesimal work done is

  $\mathrm{d}W = \vec{F} \cdot \mathrm{d}\vec{r}$

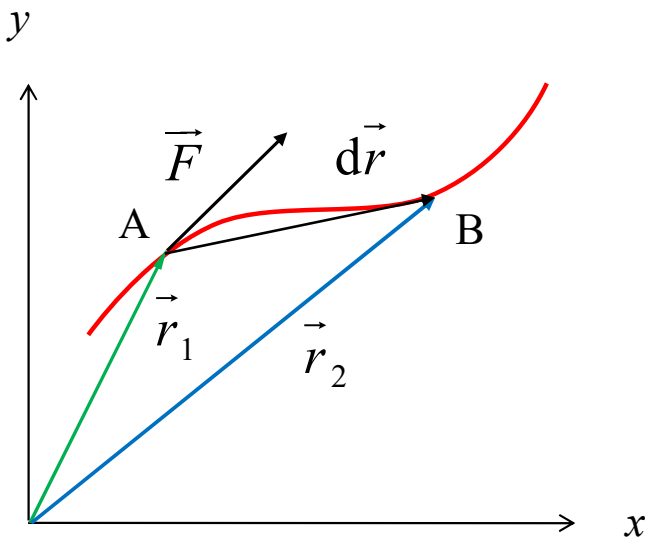
- Suppose an object moves from A to B under a variable net force.

- Since $\vec{F} = F_x \hat{i} + F_y \hat{j} + F_z \hat{k}$

  and $\mathrm{d}\vec{r} = \mathrm{d}x\,\hat{i} + \mathrm{d}y\,\hat{j} + \mathrm{d}z\,\hat{k}$

  the total work done is

  $W = \int F_x \mathrm{d}x + \int F_y \mathrm{d}y + \int F_z \mathrm{d}z$



Example 4

A force $\vec{F} = 4\hat{i}$ N acts on a particle moving along the curve $y = 4x^2$ from $x = 1$ to $x = 4$. Calculate the work done by the force.

$\vec{r} = x\hat{i} + 4x^2\hat{j}$

$d\vec{r} = dx\hat{i} + 8x dx\hat{j}$

$W = \int \vec{F} \cdot d\vec{r}$

$W = \int_1^4 4\hat{i} \cdot (dx\hat{i} + 8x dx\hat{j})$

$= \int_1^4 4\ dx$

$= [4x]_1^4 = 16 - 4$

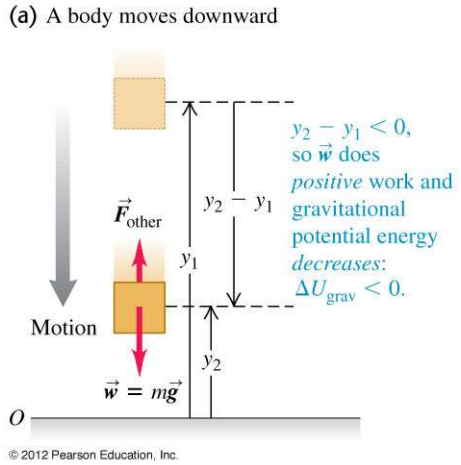
$= 12$ J

Work and Gravitational Potential Energy

- A body of mass m moves vertically down from a height y_1 above the origin to a lower height y_2.

- The work done by the constant weight is

$$W_{grav} = Fs = w(y_1 - y_2)$$
$$= mgy_1 - mgy_2$$



- We define the quantity ***mgy*** as the gravitational potential energy (referred to GPE or U_{grav}), i.e. $U_{grav} = mgy$

Work and Gravitational Potential Energy

- The work done by force of gravity is expressed as

$$W_{grav} = Fs = W(y_1 - y_2) = mgy_1 - mgy_2$$

$$W_{grav} = U_{grav,1} - U_{grav,2} = -(U_{grav,2} - U_{grav,1})$$

$$= -\Delta U_{grav}$$

- **Note that the change in GPE is relevant rather than GPE at the two points.**
- **Therefore, we can choose the reference level according to our convenience**.

Conservation of mechanical energy

- From the work energy theorem, $W_{grav} = K_2 - K_1$.
- Since $W_{\text{grav}} = -\Delta U_{\text{grav}}$

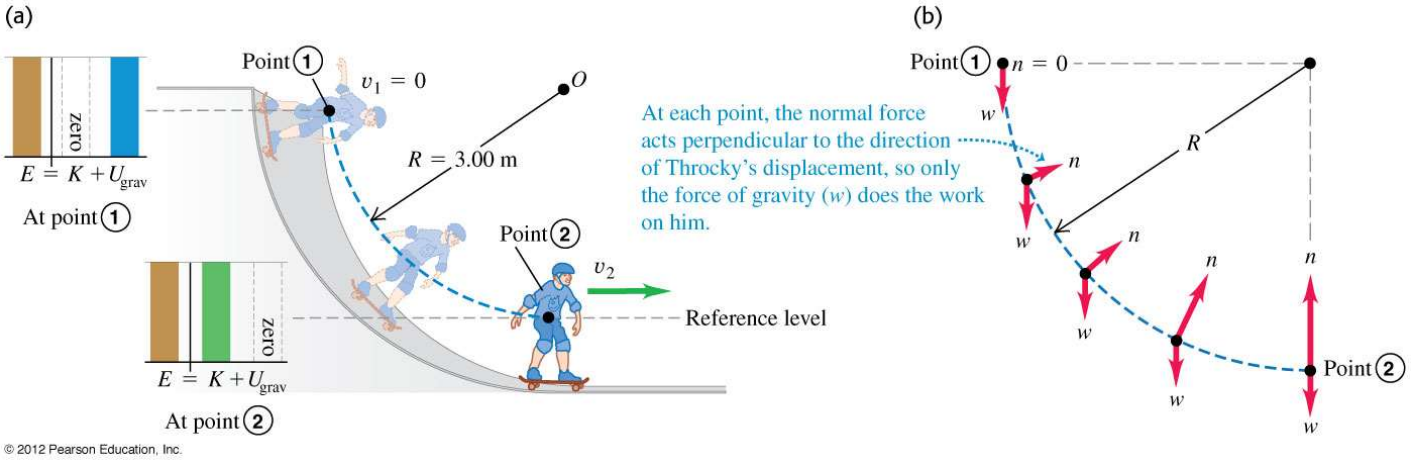
  $K_2 - K_1 = -\Delta U$

  $\Delta K + \Delta U = 0$

  Hence, $K + U = \text{constant}$
- The quantity $K + U$ is the total mechanical energy of the system.
- At any height, $K + U$ is constant.

Example 5

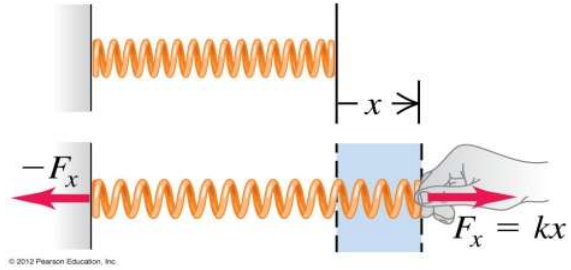
A man skateboards from rest down a curved, frictionless ramp. If we treat the man and his skateboard as a particle, he moves through a quarter circle with radius R = 3.00 m (see figure). The mass of the person and the skate board is 25.0 kg. (a) Find his speed at the bottom of the ramp and (b) Find the normal force that acts on him at the bottom of the curve.



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Elastic potential energy

- Hooke's law states that the tension in a spring is directly proportional to the elongation x of the spring, i.e. $F_x = kx$ where k is the spring constant.



- The work done by the applied force F to stretch the spring from $x = x_1$ to $x = x_2$ is

$$W_F = \int_{x_1}^{x_2} kx\,dx = \frac{1}{2}k\left[x^2\right]_{x_1}^{x_2} = \left(\frac{1}{2}kx_2^2 - \frac{1}{2}kx_1^2\right)$$

$$W_{spring} = -\int_{x_1}^{x_2} kx\,dx = \frac{1}{2}k\left[-x^2\right]_{x_1}^{x_2} = \left(\frac{1}{2}kx_1^2 - \frac{1}{2}kx_2^2\right)$$

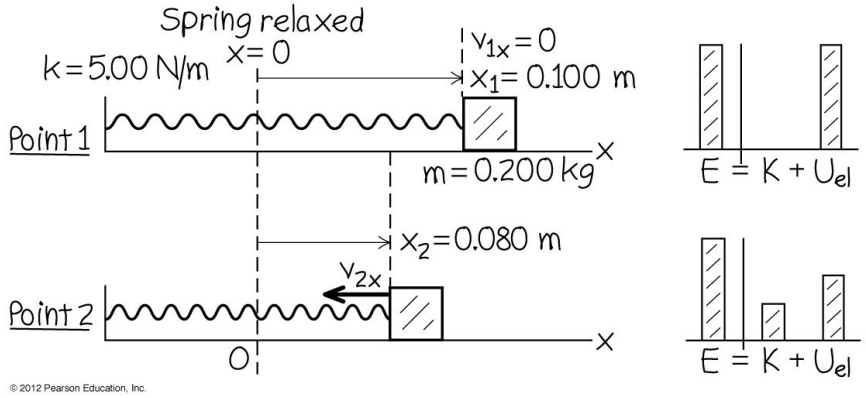
$$W_{spring} = -W_F = \left(\frac{1}{2}kx_1^2 - \frac{1}{2}kx_2^2\right)$$

$$W_{spring} = -\Delta U_{el} = -\left(\frac{1}{2}kx_2^2 - \frac{1}{2}kx_1^2\right)$$

- The term $\frac{1}{2}kx^2$ is the elastic potential energy (U_{el}) of the spring.

Example 6

A mass of 0.200 kg sits on a frictionless horizontal surface connected to a spring with force constant 5.00 N/m. The mass is pulled so that it stretches the spring 0.100 m and then released from rest. The mass moves back toward it equilibrium position. What is its x-velocity when x = 0.080 m?



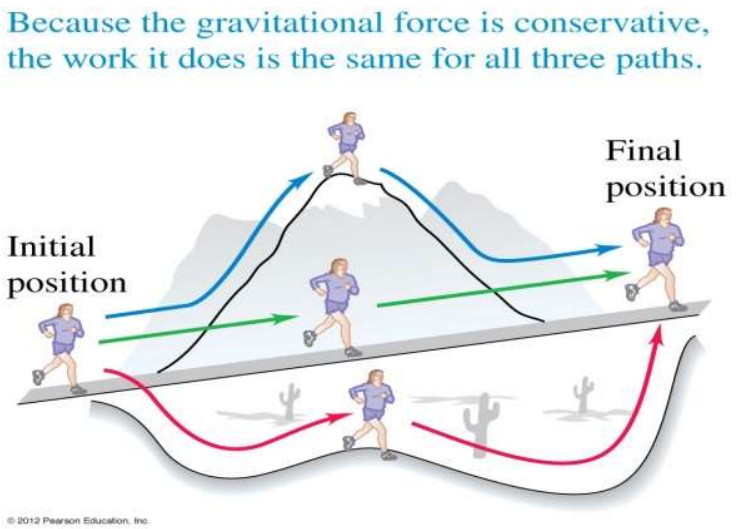
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Conservative systems and forces

- Systems where the total mechanical energy is conserved are known as conservative systems.
- The forces which act in a conservative system are known as conservative forces.
- Examples of conservative forces are gravity and tension in a spring.

Work done by a conservative force

- The work done by a conservative force can be expressed as the difference between the initial and final values of a potential energy function.
- It is independent of the path taken by the object.
- When the start and the end points are the same, the total work done by the conservative force is zero.



Conservative force and potential energy

- The work done by a conservative force is related to the change in potential energy of the system by $W = -\Delta U$.

- If Δx is the displacement of a particle, and if F_x is constant, we can write

$$W = F_x \Delta x = -\Delta U$$

$$F_x = -\frac{\Delta U}{\Delta x}$$

$$\text{In the limit } \Delta x \to 0, \quad F_x(x) = -\lim_{\Delta x \to 0} \frac{\Delta U}{\Delta x} = -\frac{\mathrm{d}U}{\mathrm{d}x}$$

- Hence we can calculate the conservative force if we know the potential energy function.

Conservative force and potential energy

- The conservative force always acts to reduce the system's potential energy.
- If the potential energy function is known we will be able to find the conservative force.
- For example the potential energy stored in a spring is $U = \frac{1}{2} kx^2$.
- The conservative force in the spring is

  $F = -\mathrm{d}U/\mathrm{d}x = -kx,$ which is the same as that given by Hooke's law.

Non-conservative force

- Examples of non-conservative forces are friction and the force unleashed in exploding fire crackers.
- The work done by a non-conservative force cannot be represented in terms of a potential energy function.
- If we toss an object onto a rough horizontal surface, it will come to rest because of work done by friction, which is converted to thermal energy of the object and the surface. i.e. $\Delta U_{\text{int}} = - W_{\text{non-conservative forces}}$
- $W_{\text{non-conservative forces}}$ is negative because friction acts opposite to the displacement of the object. Hence ΔU_{int} is positive.

Non-conservative force

- If we include the loss of the initial total mechanical energy due to non-conservative forces, we can write the law of conservation of energy as

$$K_1 + U_1 - \Delta U_{\text{int}} = K_2 + U_2$$

or $\Delta K + \Delta U + \Delta U_{\text{int}} = 0$

Power

- We often need to know how quickly the work was done.
- Therefore we define a quantity power which is the rate at which work is done.
- When ΔW amount of work is done during a time interval Δt, the average work done per unit time or the average power is

$$P_{av} = \frac{\Delta W}{\Delta t}$$

- The SI unit of power is joule per second or watt (W).
- The kilowatt (kW) and megawatt (MW) are commonly used in electricity.

Power in terms of force and velocity

- Suppose a force $\vec{F}$ acts on a body while it undergoes a displacement $\mathrm{d}\vec{s}$.
- The infinitesimal work done is $\mathrm{d}W = \vec{F} \cdot \mathrm{d}\vec{s}$
- The instantaneous rate of work done is $\dfrac{\mathrm{d}W}{\mathrm{d}t} = \vec{F} \cdot \dfrac{\mathrm{d}\vec{s}}{\mathrm{d}t} = \vec{F} \cdot \vec{v}$
- Hence instantaneous power $P = \vec{F} \cdot \vec{v}$

Example 7

A 50.0-kg marathon runner runs up the stairs to the top of Chicago's Willis tower whose height is 443 m. To lift herself to the top in 15.0 minutes, what must be her average power output?

Domestic unit of power

- The domestic unit of electricity is the kilowatt-hour (kWh).
- One kWh is the energy used by a device at a rate of 1000 watts in one hour.
- 1 kWh = 1000 W × 3600 s = 3.6×10^6 J = 3.6 MJ.
- The cost of electricity is based on the number of kilowatt-hour (kWh) of electrical energy used.

Efficiency of machines

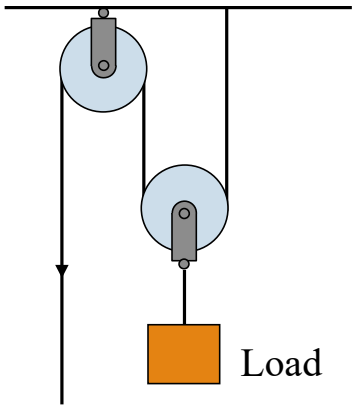
- By the principle of conversation of energy, energy input equals useful energy output plus wasted energy due to friction and other work.

$$\text{Efficiency} = \frac{\text{useful work done}}{\text{work input}} \times 100\ \%$$

$$= \frac{\text{useful energy output}}{\text{energy input}} \times 100\ \%$$

Useless work

- The useful work is the work done in raising the load.
- In the double pulley system shown below, the lower pulley and rope have weight and work must be done to raise them in order to lift the load.
- The work done to raise the lower pulley and rope are useless work.



End of chapter